

Stats 21 Course Syllabus

Python and Other Technologies for Data Science — Fall 2026

Miles Chen

2026-09-24

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Course Information

Summary

This class covers the use of Python and other technologies for data analysis and data science. The course will focus on programming with Python and a selection of its libraries: NumPy, pandas, and matplotlib for the purpose of data processing, data cleaning, data analysis, and machine learning. Other technologies covered will include Jupyter notebook and Git. Intended for Data Theory majors as an introduction to the Python language and libraries most frequently used in data science.

Learning Outcomes

Upon completion of this course, students

- will have created Jupyter Notebooks and will have published them to a Github repository
- will be able to read and write intermediate code in Python
- will have imported, manipulated, and summarized data in Python for exploratory data analysis
- will have created data visualizations in Python

Prerequisites

Stats 20

Lecture Times

- Mondays, Wednesdays, Fridays 10AM - 10:50AM
- Geology Building 3656
- Lectures will be recorded

Office Hours Schedule

- Physical Office Location: Math Sciences Building 8105 J (8th floor)
- Mondays 12:00 PM - 12:50 PM (in-person)
- Mondays 4:30 PM - 5:15 PM (online)
- Tuesdays 4:30 PM - 5:15 PM (online)
- Wednesdays 12:00 PM - 12:50 PM (in-person)
- Wednesdays 4:30 PM - 5:15 PM (online)
- Thursdays 4:30 PM - 5:15 PM (online)
- Zoom link:
 - <https://ucla.zoom.us/j/98887074323?pwd=bjFqd2ZCNnRqQmMvU3dnYkVad0RqQT09>
 - Meeting ID: 988 8707 4323
 - Passcode: 916280
 - Please **turn on your cameras**
- Also general office hours (Open to students in any class and to former students):
- Saturdays 10AM-11AM
 - <https://ucla.zoom.us/j/96806536111>
 - Meeting ID: 968 0653 6111
 - Passcode: 023194
 - Note: Office Hours on Saturday use a different Zoom link

Special Dates

Dates are subject to changes and updates. The professor will make an announcement on Canvas/BruinLearn with any changes not listed here.

Fall Quarter Dates

- Instruction begins: Thu Sep 24, 2026
- First lecture for this course: Fri Sep 25, 2026
- Week 1 begins: Mon Sep 28, 2026
- Instruction ends: Fri Dec 4, 2026
- Finals Week: Mon Dec 7 - Fri Dec 11, 2026
- Final exam for this course: Tue Dec 8, 2026, 8:00 AM - 11:00 AM

University dates follow the [UCLA academic calendar](#).

Dates with No Office Hours

- No office hours on the following dates:
 - Wed Nov 11 - Veterans Day Holiday
 - Thu Nov 26 - Thanksgiving Holiday
- No office hours during Finals Week (Dec 7-11).
- Last office hours for the quarter: Sat Dec 5 10AM
- Additional office hours cancellations will be announced on Canvas/BruinLearn.

Online Lecture

- Wed Nov 25 - The lecture before Thanksgiving will be held online. There will be no in-person class meeting.

Dates with No Class

- Wed Nov 11 - Veterans Day Holiday
- Thu Nov 26 - Fri Nov 27 - Thanksgiving Holiday

Outline of topics

- 1) Week 1: Git, GitHub, Jupyter
- 2) Week 2: Python Basics: variables, expressions, statements
- 3) Week 3: Python Functions: encapsulation, recursion, return values, iteration
- 4) Week 4: Python Data types: strings, lists, dictionaries, tuples
- 5) Week 5: Python OOP: Classes and objects, methods, inheritance
- 6) Week 6: Pythonic code: List comprehensions, kwargs; Numpy
- 7) Week 7: Pandas: Importing, reshaping, and cleaning data
- 8) Week 8: Pandas: Wrangling, and Aggregation
- 9) Week 9: Data Visualization
- 10) Week 10: Miscellaneous topics

Websites

Announcements and assignments will be posted via Canvas / Bruin Learn <https://bruinlearn.ucla.edu>
Notes will be uploaded to GitHub Repository: <https://github.com/smileschen/26-fall-stats-21>

Homework assignments will be submitted via Gradescope. Students will be invited and will need to create an account. <https://www.gradescope.com>

Software

Students will need to install git: <https://git-scm.com/downloads>

In this course, we will use Python

Books

Think Python, 3rd Edition

- By: Allen B. Downey
- Publisher: Green Tea Press and O'Reilly Media, Inc.

The textbook is available free to all at the publisher's website: <https://greenteapress.com/wp/think-python-3rd-edition/>

The print version is also available for purchase at any retailer of your choice. If you decide to purchase the text, you must get the second edition.

This is an excellent text that emphasizes brevity. Please be sure to do your reading. Every sentence is valuable.

Python Data Science Handbook, 2nd Edition

- By: Jake VanderPlas
- Publisher: O'Reilly Media, Inc.

UCLA students can read the textbook in their web browser at the following site:

<https://learning.oreilly.com/library/view/python-data-science/9781098121211/?ar>

Select "University of California, Los Angeles" and choose "Sign In with my Institution"

The print version is also available for purchase at any retailer of your choice.

Grading Policies

Grade Breakdown

- 15% Lecture Participation
- 20% DataCamp Homework
- 20% Homework Assignments
- 45% Final Exam

Letter Grade cutoffs

Letter grades are assigned on a straight scale as follows:

- 59.9 and below: F
- 60.0 - 69.9: D
- 70.0 - 76.9: C, 77.0 - 79.9: C+
- 80.0 - 82.9: B-, 83.0 - 86.9: B, 87.0 - 89.9: B+
- 90.0 - 92.9: A-, 93.0 and up A, top 5% of students: A+

I do not curve grades. I do not assign C-, D+ or D-. The top 5% of students will get an A+. An A+ is the only grade assignment that is assigned based on performance relative to peers.

If you are a Stats major or minor it is important that you take the class for letter grade. Taking the grade as pass/no-pass will not fulfill the requirements for completing the major or minor.

Lecture Participation

Lecture participation is worth **15% of the course grade**. We will use **iClicker** for two short questions about the course material during each lecture. The two questions together constitute one lecture participation quiz.

Getting Started with iClicker

iClicker is provided by UCLA at no cost to students. You may respond using the iClicker Student mobile app or the web app on a laptop; you do not need to purchase a physical clicker.

1. Follow the student account instructions on [UCLA's iClicker page](#). Sign in through UCLA's campus portal using your UCLA Logon credentials, and create or link your iClicker account as directed.
2. Confirm that the Fall 2026 Stats 21 course appears in your iClicker course list after the Bruin Learn roster is synchronized. Follow any additional enrollment instructions posted on Bruin Learn. If the course does not appear, resolve the account or enrollment issue before graded polling begins.
3. Bring a charged phone, tablet, or laptop with internet access to lecture. Open iClicker, select our course, and join the active class session.
4. Submit your own answer while each poll is open and check that your response has been recorded. Review your participation scores in Bruin Learn regularly.

Participation Credit

Each lecture quiz is worth **2 points: 1 point for responding to each question, regardless of whether the answer is correct**. Answering both questions earns full credit; answering one earns half credit; answering neither earns zero. Each lecture carries the same weight.

You must attend the lecture in person and submit your own responses during class. For the scheduled online lecture on **Wednesday, November 25**, you must attend online and respond during the live lecture. Polls close during class. Watching a recording afterward does not earn participation credit.

Answer independently unless the instructor explicitly invites discussion. Submitting responses for another student, having someone respond for you, or responding remotely to an in-person lecture is academic dishonesty. Do not use AI or other people to supply your individual responses.

Missed Quizzes

I will automatically drop your **three lowest lecture participation quiz scores**, including zeros for missed lectures, with **no questions asked and no documentation required**. Each drop applies to one full lecture's quiz, not to one individual question.

These three drops are intended to cover illness, emergencies, and other unavoidable absences. They also cover missed polls, lateness, forgotten devices, and individual technical problems. Do not treat the drops as additional absences to use in advance of an emergency.

There are no make-up quizzes, late submissions, alternative participation assignments, or additional discretionary drops. Once the three lowest scores have been dropped, all remaining lecture quiz scores count toward your participation grade. Please do not request an additional drop or a make-up for an absence or missed poll. This policy does not change approved accommodations described in the Accommodations for Students with Disabilities section.

Homework

Each homework assignment is worth the same amount. Each homework assignment will be scored out of a total of 100 points. None of the homework scores will be dropped. It is your responsibility to verify that your homework assignment successfully uploaded by the deadline.

All homework assignments will be posted as an ipynb file. You will download the ipynb file and modify it with your answers.

After entering all of your answers, select “Kernel” > “Restart and Run All”. Check to make sure all requested answers are clearly visible.

Then students will submit the PDF file to Gradescope. Do not submit the ipynb file.

Homework submission requirements

- Unless instructed otherwise, homework **must be a PDF with letter-sized pages**.
- When submitting homework to Gradescope, the **pages of the submission must be properly assigned to/tagged with the corresponding question**.
 - When assigning pages, make sure ALL of the corresponding output for a particular question is assigned to the question. You can tag multiple pages to a single question. A single page can be tagged to multiple questions.
 - For the purposes of determining if a submission is late or not, Gradescope uses the time of the upload as the submission time. You can take additional time to assign/tag the pages and your submission time will not be affected.
 - If the output is not correctly assigned to/tagged with the question, it will be graded as if the output is missing. Regrade requests due to incorrect assignment or tagging of output will not be considered.
- Output for each homework question must be clearly visible to receive credit.
 - Graders do not have the luxury of trying to run your code or debug your code. They may not even have time to see what your individual lines of code can do.
 - The graders will look for the printed results, so it is **very important** you properly print the requested results in your document. It is your job to make sure that you get the credit you deserve.
- It is very important you review your own output before submitting. Sometimes a small edit in one section will cause other sections (that used to work with no issues) to suddenly break. We cannot give you credit for output that results in errors. **Be sure to review all pages of your output before submitting.**

Example of bad submission:

Problem 1: Use a loop to produce a list of the first 10 numbers in the Fibonacci sequence

```
x = [1, 1]
for i in range(1, 9):
    x.append(x[i-1] + x[i])
```

The grader has no idea what was produced. You will receive 0 points for this submission.

Example of good submission:

Problem 1: Use a loop to produce a list of the first 10 numbers in the Fibonacci sequence

```
x = [1, 1]
for i in range(1, 9):
    x.append(x[i-1] + x[i])
print(x)

[1, 1, 2, 3, 5, 8, 13, 21, 34, 55]
```

This is the exact same code, but adds the line `print(x)`. This small change makes a big difference.

It is your responsibility to read through the rendered output and to make sure that every problem and task has the required output before submitting your assignment.

Late Policy for Homework

Homework deadlines are based on local time in Los Angeles, California.

There is a 10 minute grace period for homework submissions. Submissions up to 10 minutes and 59 seconds late will be accepted with no penalty.

Homework assignments submitted 11 minutes late or more will be accepted with penalty. There is a deduction of 5 points for being less than one hour late. An additional 5-point deduction will be taken for each additional hour it is late.

Deductions are suspended between the hours of 12 am and 5:59 pm the following day. The penalties resume at 6 pm.

The maximum late penalty is 60 points.

So if an assignment is due at 6pm, a homework assignment submitted on Gradescope at

- 6:10:59 pm will receive no penalty.
- 6:11:00 pm will receive 5-point penalty.
- 6:59:59 pm will receive 5-point penalty.
- 7:00:00 pm will receive 10-point penalty.
- 7:59:59 pm will receive 10-point penalty.
- 8:00:00 pm will receive 15-point penalty.
- 11:59:59 pm will receive 30-point penalty.
- 12:00:01 am (following day) will receive 30-point penalty.

- 5:59:59 pm (following day) will receive 30-point penalty.
- 6:00:00 pm (following day) will receive 35-point penalty.
- 10:59:59 pm (following day) will receive 55-point penalty.
- 11:00:00 pm (following day) will receive 60-point penalty.
- Anything later (but less than 7 days late) will be accepted with a 60-point penalty.

I strongly advise uploading the homework to Gradescope well in advance of the deadline in case there are connectivity problems or server issues.

After 7 calendar days, late homework assignments will not be accepted. Exceptions beyond the 7-day submission window will be granted in only extreme circumstances and will only be granted via an office hours meeting (Zoom or in-person) with the professor.

Homework extensions

I generally grant a 72-hour reprieve from the late penalty for homework submissions if you contact me in office hours to request an extension prior to the homework deadline. Extension requests made via email will be ignored.

Requests for extensions after the homework deadline will be granted only in extreme situations.

Homework Regrades

Homework assignments are graded by TAs. Occasionally the grader may make a mistake in grading your submission.

Both the TA and the professor have the authority to change the grade of a homework assignment.

All requests for regrading a homework assignment must be submitted via Gradescope.

Do not email the TA or the professor for a request to regrade an assignment.

Please only submit a regrade request if you are certain that a mistake was made in the grading of your homework.

During a regrade request, the TA or Professor may discover and deduct points for mistakes that were not found during the initial grading. It is possible your regrade request results in a lower grade for the assignment.

DataCamp Homework

Some homework assignments will be assigned via DataCamp.

Students will receive an invitation to sign up for a free 6-month subscription to DataCamp. The email address that appears on Canvas and my.ucla will be used for the invitation.

Students **must** sign up for DataCamp using the **exact same** email address that was invited. For example, if joebruin@ucla.edu is invited, joebruin@g.ucla.edu CANNOT be used.

Students must also make sure their DataCamp account uses the first and last name as it appears in Canvas.

Each homework assignment on DataCamp will consist of completing an assigned course. Each homework assignment is worth the same amount.

Tentative Homework Deadlines

Homework deadlines subject to change. Additional DataCamp courses could be assigned.

Regular Homework

- HW1: Fri October 9, 2026
- HW2: Wed October 21, 2026
- HW3: Wed November 4, 2026
- HW4: Fri November 20, 2026
- HW5: Wed December 2, 2026

DataCamp Homework

- Intro to Shell: Fri October 2, 2026
- Intro to Git: Wed October 7, 2026
- Intro to Python: Mon October 12, 2026
- Intermediate Python: Mon October 19, 2026
- Data Manipulation with Pandas: Mon October 26, 2026
- Joining Data with Pandas: Mon November 2, 2026

Final Exam

There will be a final exam covering programming concepts in Python.

The final exam is scheduled for **Tuesday, December 8, 2026, 8:00 AM - 11:00 AM** (Los Angeles time). See the [UCLA Schedule of Classes](#) for the exam location and any updates.

Make up policy for Exam

If you know you will be unavailable prior to the exam, please visit me in office hours to explain the situation. I will work with you to find an alternative arrangement.

If you are ill on the day of the exam, stay home and do not come to class. Be sure to message me within 24 hours of the exam and attend the next available office hours to arrange for a make-up exam.

Other emergent situations for missing the exam will require documentation.

In most cases, make-up of the exam must be taken within 7 days of the actual exam, or a zero will be recorded. (For example, a missed exam originally given on Monday must be made up by the following Monday by 5pm.)

Course Grade Changes

After course grades have been submitted to the registrar, grades are final. Grade changes will only be considered if there has been a clerical or procedural mistake. Students have one quarter to make requests for a grade change. Graded exams and other materials will be kept for one quarter. After one quarter, course grade changes will not be made.

Other Course Policies

Some of the following verbiage has been taken and adapted from the CEILS Syllabus Template as well as various campus resource websites.

Copyright Policy

All course materials posted on the course website or distributed in class (including but not limited to lecture slides, homework assignments, lecture videos, quizzes, exams) are intended for personal use only by students enrolled in this course. **It is a violation of course policy to post, share, or distribute any course material electronically or physically without permission from the instructor, even after the course is over.**

Policies on Academic Integrity

As a student and member of the University community, you are here to get an education and are, therefore, expected to demonstrate integrity in your academic endeavors. All students must uphold University of California Standards of Student Conduct as administered by the Office of the Dean of Students. Students are subject to disciplinary action for several types of misconduct, including but not limited to: cheating, multiple submissions, plagiarism, prohibited collaboration, facilitating academic dishonesty, or knowingly furnishing false information.

In addition, each student is the sole owner of their own code and work and must NOT do any of the following:

- Submit work that was written by someone other than the student.
- Use an AI tool (including but not limited to ChatGPT, Github Copilot, Gemini, Claude) to write solutions.
- Publish code or solutions online.
- Post the course questions on forums other than the designated course discussion forum (Campuswire).
- Submit someone else's work or a modification of that work, with or without that person's knowledge.
- Allow someone else to submit their work or a modification of their work.
- Contract course work out to others.
- Plan or execute with another student(s) some form of cheating during an exam.
- Make use of unauthorized material during an exam or for preparation of an exam.
- View completed assignments or exams from the current or previous iterations of the course, regardless of their source.

Often, a student who considers cheating feels unprepared and under pressure. If you are at a point where you are contemplating cheating, please come talk to me instead.

For more information about academic integrity and student conduct code, please go to <http://www.deanofstudents.ucla.edu/Student-Conduct>

Sexual Harassment, Sexual Violence, and Gender Discrimination

Title IX prohibits sexual harassment, sexual violence, and any discrimination based on gender, gender identity, sexual orientation, pregnancy.

The University of California is committed to creating and maintaining a community dedicated to the advancement, application and transmission of knowledge and creative endeavors through academic excellence, where all individuals who participate in University programs and activities can work and learn together in an atmosphere free of harassment, exploitation, or intimidation.

If you, or someone you know, have been a victim of sexual harassment or sexual violence, please know that you are not alone and you can seek help. The University offers many resources to aid students and other victims.

The Title IX office offers both confidential and non-confidential resources for survivors.

<http://sexualharassment.ucla.edu/>

<https://sexualharassment.ucla.edu/resources/resources-for-complainants>

Any type of harassment of any other student will not be tolerated.

Please be aware that your TA and Lecturer are legally compelled to report any information in regards to sexual harassment or sexual violence to the Title IX officer.

Confidential support and advocacy is available for students who have experienced sexual harassment or sexual violence at the CARE Advocacy Office for Sexual and Gender-Based Violence, John Wooden Center West 1st Floor, (310) 206-2465, advocate@careprogram.ucla.edu. Sexual violence or sexual harassment can also be reported directly to the University's Title IX Coordinator, 2241 Murphy Hall, titleix@conet.ucla.edu, (310) 206-3417.

Reach a confidential UCLA advocate, 24 hours a day, 7 days a week by calling the CAPS 24/7 Crisis Hotline (310) 825-0768

Mental Health and Student Well-Being

Counseling and Psychological Services (CAPS) is here to support your mental health needs as you pursue your academic goals.

All registered UCLA students are eligible for same-day clinical consultation and help with referrals, linkage, treatment recommendations, and coordination of mental health care and support regardless of insurance plan or current residence.

24/7 crisis support is always available by phone at (310) 825-0768.

<http://counseling.ucla.edu>

Attending and completing university coursework is difficult and can become burdensome both physically and psychologically. Take care of your mental well-being and seek support from some of the many excellent resources offered at CAPS.

It's every Bruin's responsibility to refer or reach out on behalf of students who are in distress or might harm themselves or others. If you are concerned about another student or member of the campus, please consider consulting with a professional.

<https://counseling.ucla.edu/concerned-about-a-bruin/concerned-about-a-bruin>

Accommodations for Students with Disabilities

Students needing academic accommodations based on a disability must contact the Center for Accessible Education (CAE) at (310) 825-1501 or present in person at Murphy Hall A255. As the professionals delegated authority from the campus to determine reasonable disability accommodations, CAE will assess all requested accommodations and communicate appropriately with faculty. In the event that a student has approval for proctoring arrangements during exams, please inform your respective professors and/or Teaching Assistant before date of exams. When possible, students should contact the CAE within the first two weeks of the term as reasonable notice is needed to coordinate accommodations.

For more information visit <http://cae.ucla.edu>

General Syllabus Policy

The instructor reserves the right to make changes to the syllabus. All changes to the syllabus will be announced to the class.

If any policy in the syllabus requires interpretation, the instructor will clarify the policy and the instructor's interpretation of the policy is final.

Additional Information

Undergraduate Mentoring

I am a mentor for undergraduates for the UCLA Department of Statistics. This means that you may visit me during office hours (or by appointment) for an informal meeting where you can ask questions unrelated to course content, such as research opportunities, graduate studies, career paths, or any other topic pertinent to your education. You are welcome to visit and meet even after the course ends.

Requests for Letters of Recommendation

If you think you might request a letter of recommendation in the future, please make an effort to make yourself known to me. I am not able to write strong letters of recommendation for students I do not know well.

I also do not have a good memory, so please visit office hours frequently and repeatedly introduce yourself to me. Please attend class in person and sit near the front of the classroom where I will see and recognize your face. Actively participate in class.

For a full guideline, please see this page on my personal website: <https://smileschen.com/lor.html>

Syllabus Copyright

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